

Folder	Paper	Claim	Tier	Falsifier / Kill-Cond
Predictions				
Gravity	031	$a_0(z) = cH(z)/(2\pi)$ evolves factor-for-factor with H	Prediction	departs from $cH(z)/2\pi$. Only 031 §8.8 is data-confronted: MUSE-DARK III (A&A 2026) $\sim 30\sigma$ evolution, $\sim 8\%$ local, edge of $H(z)$; $\alpha=1$ the decisive open number. Do NOT over-bank (direction won, exact rate tense)
Gravity	029	$a_0 = cH_0/(2\pi) \approx 1.08 \times 10^{-10} \text{ m/s}^2$, local MOND transition acceleration (the redshift evolution 1.15 is the forward weapon)	Derived	a_0 tracks H_0 wrong as the Hubble tension resolves; wide binaries at a_0 obey plain Newton
Gravity	GR-IV	α_1, α_2 preferred-frame (the keystone kill-switch)	Prediction	ANY nonzero α_2 kills it ($\alpha_2 = 0$ EXACT, structural); $\alpha_1 > \sim 1e-5$ kills it ($\alpha_1 < \sim 4e-93$ safe). Passed favorably
Gravity	OffsetVelocity/117	Offset-velocity knee law: flat-knee-line-ceiling	Prediction	Λ CDM velocity-independent scatter, or MOND-EFE smooth roll-off, refutes. Only the SHAPE is ingredient-robust; the knee location rests on a flagged $\nu \sim 0.70$ (3D O(3)) assumption
Cosmology	DF2/DF4	Group-embedded low-accel galaxies show a KNEE-like suppression sharper than MOND-EFE; void UDGs full enhancement	Prediction	Preds 1-4: environmental independence (Λ CDM) or smooth MOND-EFE roll-off; knee sharper than MOND (Pred-4 uniquely ED, below current precision)
Dark-sector	DarkSector	Dark matter (relic) is WARM ($\sim 0.2\text{--}0.7 \text{ keV}$), not a cold WIMP	Prediction	F1: a cold heavy WIMP detection, or perfectly-cold DM, refutes the warm relic. LIVE RISK: mechanism prefers cold GeV-Planck (6-26 OOM too heavy)
Dark-sector	DarkSector	Galaxies contain NO dark-matter halo at all	Prediction	F2: DM substance found INSIDE galaxies (halo, CDM cusps, scattered mass) refutes the horizon-only galactic account
Gravity	038.6	Activity-dependent weak lensing $\sim \text{grad } \mu ^2$	Prediction	activity-INDEPENDENT lensing refutes. PROVISIONAL — $\alpha_{\text{act}}, A(x)$, the $ \text{grad } \mu ^2$ -vs-feedback discriminator observable all OPEN
Cosmology	Cos-05	Dark energy strictly $w=-1$, no time-variation (vs quintessence/phantom)	Prediction	F1/F4: clean phantom $w < -1$ over wide z , or $w\text{-dot} \neq 0$, or $\Lambda\text{-dot} \neq 0$. Distinctive $w=-1$ null; ties 22-Ways $w=-1$
Dynamics	Dyn-02	EoS confined to $\{-1, 0, 1/3\}$ + composites (no clean phantom $w < -1$, no stiff $w=1$)	Prediction	a clean phantom $w < -1$, or a stiff $w=1$, outside the substrate-admitted $\{-1, 0, 1/3\}$ +composites set refutes it. NEAR-distinctive, but sits BEHIND the Q1/Q2 OPEN (currently inherited, not derived); ties to Cos-05 $w=-1$
Q-compute	056	Class-A wall EXISTS as sharp + architecture-independent (the shape is the claim, not the number)	Prediction	no sharp architecture-independent wall (smooth environment/architecture-DEPENDENT decoherence) refutes it. Standard decoherence-only + CSL predict no such wall. NUMBER 140-250 kDa is PRESSURED (≥ 170 kDa interference already shown, Nature 2026, still QM-consistent) + the second-harmonic leg CONFOUNDED (KDTL optics); the SHAPE is the claim -- do NOT over-bank the number
Q-compute	058b	Class-C plateau, RE-SCOPED to broadcast-type redundancy (repetition / GHZ / cat), NOT surface-code distance	Prediction	surface-code clean suppression (Willow $\Lambda \sim 2$) is consistent-BY-CONSTRUCTION, NOT a passed test. Live pair = repetition-code floor persistence (Google $\sim 1e-10$ above $d \sim 15$) + cat-state width ceiling; a floor surviving burst-mitigation at cross-platform-consistent content = the ED signature. Standard FT predicts error > 0
Substrate-evaluation	CleanSubstrateVector	Parity violation is a spontaneous, contingent orientation correlated with the matter/antimatter sign	Prediction	a demonstrated LAW-LEVEL (non-spontaneous, non-contingent, fixed) origin of the weak handedness refutes it. Distinctive vs the SM's law-level fixed handedness
predictions	ED-09.5	Slow envelope modulation at bifurcation, $f_v \approx 1.1 \cdot \nu_{\text{dec}}$ (the Q/π form), tested across 10+ orders of magnitude in decoherence rate	Prediction	No envelope detected at the predicted frequency band ($\sim 11\text{--}110 \text{ Hz}$) in any of 3+ uncorrelated datasets (optomechanics + FRAP residuals)
qft	MS-II	No parity-violating abelian $U(1)$ force; the abelian coupling is chirality-blind by construction	Prediction	Discovery of a chiral (parity-violating) $U(1)$ gauge force
qft	MS-II	No stable fundamental gauge group larger than $SU(3)$; stable channel families are exactly $\{1..d\}$ with $d=3$	Prediction	Discovery of a stable fundamental $SU(N \geq 4)$ sector
Theory				
Primitives	P01	Existence layer (the foundational relational existence; below it, no further substrate structure)	Primitive	
Primitives	P02	Participation as the primitive relation (chains participate — derived composite structures)	Primitive	
Primitives	P03	Channel + locus indexing; spatial homogeneity (bundles three commitments, splittable)	Primitive	
Primitives	P04	Bandwidth — a non-negative additive scalar	Primitive	
Primitives	P05	Polarity-transport between adjacent loci	Primitive	
Primitives	P06	Spatial dimension $D = 3+1$	Primitive	
Primitives	P07	Channel structure as an ontological primitive	Primitive	
Primitives	P08	Substrate scale I_{ED} ($= I_{\text{P}}$ empirically)	Primitive	
Primitives	P09	$U(1)$ -valued polarity	Primitive	
Primitives	P10	Rule-type primitive (licenses V5 as a P10 kernel rule-type)	Primitive	

Primitives	P11	Commitment — irreversible, with environmental phase-randomization (THE arrow primitive)	Primitive
Primitives	P12	Stability landscape $\Sigma = \text{Coh} - \text{Str} - \text{Grad}$	Primitive
Primitives	P13	Time homogeneity; primitive event-discreteness (NOT 'the tick')	Primitive
Constants-inherited	Paper_012 (P-RB-1)	c — the rate of becoming (tick-to-hop conversion)	Constant
Constants-inherited	Paper_027	$G = c^3 \cdot I_P^2 / \hbar$ — Newton's constant	Constant*
Constants-inherited	Paper_029	$a_0 = \hbar c / (2\pi)$ — the MOND transition scale	Constant*
Constants-inherited	Paper_038_5	Λ — the cosmological constant	Constant*
Constants-inherited	Paper_087	$\hbar$ — the action of one commitment (the grain of memory)	Constant
Constants-inherited	Paper_087 (P08) / 089	$I_P = I_{ED}$ — the grain (size of one hop)	Constant
Constants-inherited	cosmology	H_0 — the Hubble rate	Constant*
Constants-inherited	matter sector	Particle masses	Constant
Constants-inherited	matter sector	Gauge couplings g	Constant
Constants-inherited	matter sector	$\alpha \sim 1/137$ (fine-structure); mass ratios	Constant
Cosmology	SCBU	a_0 & $\chi_{\text{canonical}}$ FORCED to share one substrate-cosmology boundary origin	Postulated
Entanglement	063	E-1 factorizable $\Psi^{AB} = \Psi^A \times \Psi^B$ & the tensor-product structure	Postulated
Entanglement	064	V5-bilocal has Schmidt rank > 1 generically	Postulated
Entanglement	065	Monogamy budget cap; measure saturates the budget	Postulated
Entanglement	069	ED sits at Tsirelson; PR-boxes excluded	Postulated
Gravity	035	Six acoustic-metric guardrails (C1-C6)	Postulated
Gravity	QuadraticStrain	P-Quadratic-Strain (Model C)	Postulated
QFT	019	P-YM-Action-Coarse-Graining	Postulated
QFT	020	P-OS-Reflection-Positivity (OS3)	Postulated
QFT	021	P-Profile-Rescaling (YM continuum survival)	Postulated
QM-kinematics	002	Tensor-product composition $\Psi^{AB} = \Psi^A \times \Psi^B$	Postulated
QM-kinematics	012	P-RB-1 local rate of becoming; substrate-c constancy	Postulated
Relativistic-QM	114	P-Gauge (paper-specific postulate)	Postulated
Substrate-evaluation	QuantumDarwinism	P-QD-LiveWeight (uncommitted branch-basis correlation consumes pairwise V5 weight)	Postulated
Substrate-evaluation	V5AttractiveSign/RelationalTick/MassWithoutMass	V5 existence as a cross-chain kernel	Postulated

Claims

Black-hole	039	Horizon = $b \rightarrow 0$ decoupling surface (Γ_{cross} collapse)	D-via-I / Form-forced
Black-hole	040	Trans-Planckian resolution (modes terminate at w_c)	D-via-I / Form-forced
Black-hole	041	Planck-mass remnant $M^* = c \cdot I_P$ (Scenario C)	D-via-I / Form-forced
Black-hole	042	No-singularity (C_{cum} bounded, $r \geq I_P$)	D-via-I / Form-forced
Black-hole	043	Area-law $\text{FORM } S \propto A$	D-via-I / Form-forced
Black-hole	044	BHPT scattering / Regge-Wheeler form	D-via-I / Form-forced
Black-hole	045	Helicity preserved (Schwarzschild) / amplified (Kerr)	D-via-I / Form-forced
Black-hole	046	Kerr twist $g_{t\phi}$	D-via-I / Form-forced
Black-hole	047	Hawking temperature $T_H = \kappa / 2\pi$ via V5 KMS	D-via-I / Form-forced
Black-hole	047	Non-thermal Hawking correction $[1 - c_1(w/w_c)^2]$	D-via-I / Form-forced
Black-hole	048	Scenario C forced (A, B excluded)	D-via-I / Form-forced
Black-hole	049	PBH relic abundance $\Omega_{\text{PBH}} \propto n_{\text{PBH}} \cdot M^*$	D-via-I / Form-forced
Black-hole	050	Page-curve rise + turnover + decline	D-via-I / Form-forced

Dark-sector	RelicLagrangian	Relic $\rho \propto a^{-3}$ from the P11 commitment number	Derived	
Dark-sector	RelicLagrangian	Sector is TWO-COMPONENT, not one substance (L_{self} coarse-grains CANONICAL)	Derived	
Dark-sector	RelicLagrangian	Pressureless-dust limit $w \sim 0$ (fast-oscillating massive scalar)	Derived	
Entanglement	064	Schmidt rank $> 1 \Rightarrow$ non-factorizable	Derived	
Entanglement	065	Monogamy (CKW-form inequality)	Derived	
Entanglement	066	No-signaling $p(a x,y)=p(a x)$, over-determined by 3 locks	Derived	
Entanglement	068	von Neumann entropy form $S = -\text{Tr } \rho \log \rho$	Derived	
Foundations	089	Theorem N1 — V1 finite-width (G bounded, decaying, scale l_{ED} ; delta-function and inf-width limits excluded)	Derived	
Foundations	089/093	Theorem T18 — V1 strictly retarded support (the kernel-level arrow); advanced-V1 refuted by P11, symmetric-V1 non-constructible	Derived	
Foundations	089/093	Wheeler-Feynman engagement: symmetric electrodynamics is mathematically consistent but substrate-operationally non-constructible in ED	Derived	
Foundations	090/094	V5 inherits retardation from V1 (via T18); primitive-forward-causality $\rightarrow$ phenomenological arrows inherit under DCGT	Derived	
Gravity	025	Holographic bound $N(R) \leq 4\pi R^2 / l_{\text{ED}}^2$ (area law as edge-count)	Derived	
Gravity	026	Newton $a_N = GM/R^2$ from the cumulative-strain reading	Derived	
Gravity	032.5	Free chains follow acoustic-metric geodesics	Derived	
Gravity	034/036	Deep-MOND $a = \sqrt{a_N a_0}$; flat-background AQUAL field eq	Derived	
Gravity	037	a_0 continuum-limit invariant (does not RG-run)	Derived	
Gravity	GR-I	Temporal lapse $N^2 \sim b \rightarrow$ weak-field Einstein metric (Schwarzschild branch, not Nordstrom)	Derived	
Gravity	GR-I	Emergent $g \sim 1/b$ + factor-of-two light bending	Derived	
Gravity	GR-II	Field-equation form $G + \Lambda g = \kappa T$	Derived	
Gravity	GR-II	Matter conservation $\text{div } T = 0 \Leftrightarrow$ geodesic motion	Derived	
Gravity	GR-II	ED gravity is KHRONOMETRIC (2 tensor + 1 khronon); gauge = foliation-preserving diffeos	Derived	
Gravity	GR-II	$c_T = c$ structurally (GW170817 as identity); universal (not differential) Lorentz violation	Derived	
Gravity	GR-III	$\alpha=1$ (P-Commitment-Linear) FORCED; Nordstrom excluded	Derived	
Gravity	GR-III	Khronon scalar speed $c_s = c$	Derived	
Gravity	GR-IV	$\alpha_2 = 0$ EXACTLY	Derived	
Gravity	GR-IV	$\alpha_1 = -4 \cdot \lambda_{\text{local}}$; commitment FORCED sparse $\Rightarrow \lambda_{\text{local}} < 1$ (≥ 70 orders safe)	Derived	
Gravity	MetricFromGraph	Reach law $p=1/(d-1) \Rightarrow g \sim 1/b$ in EXACTLY 3D	Derived	
Q-compute	054	UR-1.1/1.2/1.3 three unresolvedness statements ($M_{\text{eff}} < M_{\text{cap}}$; $\Gamma_{\text{cross}} > \Gamma_{\text{decoupling}}$; $\Gamma_{\text{commit}} < \Gamma_{\text{indiv}}^{-1}$)	Derived	
Q-compute	054/056	Existence of Γ_{commit} , $\Gamma_{\text{indiv}}^{-1} \sim c/l_{\text{ED}}$; $M_{\text{cap}} = (\Gamma_{\text{indiv}}/\Gamma_{\text{commit}})^{1/\gamma}$ as the UR-1.1 threshold	Derived	
Q-compute	058b	$\alpha_{\text{topological}} = 0$ (per-locus): the 058 §3.3 saturation never triggers for topological codes	Derived	
Q-compute	058b	Syndrome records are budget-free (P11 commitments)	Derived	
QFT	016	Continuum gauge-covariant derivative $D_\mu = \partial_\mu + iA_\mu$	Derived	
QFT	017	Lorentz covariance of the continuum free-scalar equation	Derived	
QFT	019	Continuum YM equations $D_\mu F = J$ (Euler-Lagrange)	Derived	
QFT	021	Spectral-gap FORM $\Delta \sim 1/l_*$ (Yang-Mills mass gap)	Derived	
QFT	022	Substrate quotient = physical-state space; BRST = coarse-grained quotient	Derived	
QFT	024	Lindblad/GKSL master-equation form	Derived	
QFT	MS-II	Gauge group $U(N) = \text{SU}(N) \times U(1)$ from channel multiplicity	Derived	
QM-kinematics	001	Complex polar carrier $P_K = \sqrt{b_K} \cdot e^{i\phi_K}$	Derived	
QM-kinematics	009	Berry phase γ_C = holonomy of the coarse-grained P05 connection	Derived	

QM-kinematics	010	Aharonov-Bohm phase from T17 bundle circulation	Derived	
QM-kinematics	011	Bloch form $\psi_k = e^{i k x} u_k(x)$ from discrete translation symmetry	Derived	
QM-kinematics	012.7	Galilean adjacency/propagation bandwidth-asymmetry	Derived	
Relativistic-QM	103	CI(3,1) frame uniqueness up to similarity	Derived	
Relativistic-QM	104	Anyon prohibition in 3+1D	Derived	
Substrate-evaluation	A1 (CommonCauseNotChannel)	Controlled channel capacity between distinct regions is EXACTLY zero; boundary = common-cause not channel	Derived	
Substrate-evaluation	B4 (ChargeAsTopology)	Winding w in Z conserved + irreversibility-protected; orientation-blindness realizes the skeleton, withholds the field	Derived	
Substrate-evaluation	CleanSubstrateVector	The parity-clean substrate is VECTOR for every channel-count ($\det H(k)$ even $\Rightarrow$ winding=0 for all N); parity violation NECESSARILY spontaneous	Derived	
Substrate-evaluation	QuantumDarwinism	Committed records cost no V5 budget (accounting theorem) $\Rightarrow$ classical objectivity unbounded	Derived	
Constants-inherited	Constants-paper §1	Constants ledger: $c/\hbar/l_P$ present at first events (dimensionful, ~units); $a_0/\Lambda/H_0$ /masses inherited late (dimensionless, the personality)	Grounded	
Constants-inherited	Constants-paper §3	$G = c^3 \cdot l_P^2/\hbar$; G absent from the SM; $[G_d] = m^d$ (units know the dimension count)	Grounded	
Constants-inherited	Constants-paper §4	The dimension-signature in $G(l_P^{d-1})$ IS the holographic area-law / reach-law road — NOT a new independent road to 3D	Grounded	
Cosmology	Cos-01	Inflation = post-SCBU substrate SATURATION regime; $a(t)=a_0 e^{Ht}$	Grounded	
Cosmology	Cos-05	Dark energy = late-universe SATURATION regime; $w=-1$ (leading order)	Grounded	
Cosmology	Cos-05	Λ -smallness (120-OOM) reduces to Route A + Friedmann; naive $\rho_{vac} \sim M_P^4$ STRUCTURALLY ABSENT substrate-side (no infinite mode-tower)	Grounded	
Cosmology	DF2/DF4	Low-accel galaxies CONSISTENT WITH (not counterexamples to) derived a_0 , via commitment-density saturation suppression	Grounded	
Cosmology	SCBU	a_0 & $\chi_{i,canonical}$ = two projection MECHANISMS of one boundary $R_H=c/H_0$	Grounded	
Dark-sector	DarkSector	Committed NEUTRAL RELIC candidate for clusters/CMB	Grounded	
Dark-sector	RelicLagrangian	Field content $\Phi = \sqrt{\rho_0/m_R} e^{i\theta}$ = coarse-grained neutral-channel participation measure	Grounded	
Dark-sector	RelicLagrangian	Chronometric kinetic form L_{kin} (time-along- u + spatial c_{perp})	Grounded	
Dynamics	Dyn-01	Saddle-Hessian-signature dynamics = the foundational generator; 3 regime classes (S1 collapse / S3 inflation-DE / S2 horizon-radiation)	Grounded	
Dynamics	Dyn-02	Horizon-motion law: RDE($w=1/3$), MDE($w=0$), LDE($w=-1$) via substrate Noether stress-energy + DCGT + Friedmann	Grounded	
Dynamics	Dyn-03	Radiation law: EM Larmor + GW quadrupole via Noether flux + DCGT	Grounded	
Dynamics	Dyn-04	Gravitational collapse = monotonic Hessian-signature evolution toward all-compression; horizon = closed signature-class boundary	Grounded	
Dynamics	Dyn-05	Compact-binary inspiral = two-saddle coupled evolution + quadrupole radiation reaction; Peters timescale	Grounded	
Dynamics	GW-00	GW = propagating saddle-Hessian-signature reconfiguration (V1 retarded); energy = local ED-gradient load (no pseudotensor)	Grounded	
Dynamics	GW-00	Quadrupole source-amplitude FORM $P_{GW} \sim \langle Q_{\text{triple}} \cdot \dot{}^2 \rangle$	Grounded	
Dynamics	GW-01	BH ringdown = decay of a near-horizon mixed-signature saddle; mode quantization from Hessian eigenstructure	Grounded	
Dynamics	GW-02	Stochastic background $\Omega_{GW}(f)$ = cosmologically-integrated NoetherFlux over saddle transitions	Grounded	
Entanglement	063	E-2 bilocal cross-chain term Δ_{KL} exists	Grounded	
Gravity	032	Six ED-10 weak-field prerequisites fulfilled at form level	Grounded	
Gravity	033	Scalar-tensor acoustic-metric field eq (covariantization)	Grounded	
Gravity	038.5	$\Lambda = V_1$ cosmological-scale integral over R_H ; Λ CONSTANT, $w=-1$ forced, = frozen saturation-floor	Grounded	

Gravity	KM-I	The khronon's deep field IS the MOND sector (unifies the two gravity programs)	Grounded	
Gravity	KM-I	Lensing tracks the MOND potential, no added vector; no new ghost; PPN-safe	Grounded	
Gravity	KM-II	Orthogonality theorem: $\theta=0$ in statics $\Rightarrow$ the regulator family is invisible to every tested result	Grounded	
Gravity	KM-II	Clustering dial = the soft spot (one dial sets degeneracy-cure + Cerenkov + clusters/CMB)	Grounded	
Gravity	KM-II	One-Lambda thesis: $W0 = -24 \cdot \pi^2 \cdot \Omega_{\Lambda}$ ~ -162	Grounded	
Q-compute	053	$M_{\text{cap}} = \min(N_{\text{bw}}, N_{\text{V5}}, N_{\text{commit}})$ is one substrate object; min-structure forced by simultaneity	Grounded	
Q-compute	054	The three UR-1 rate quantities have distinct PRIMARY DRIVERS (basis of A/B/C distinctness)	Grounded	
Q-compute	055	A/B/C three-class EXHAUSTIVENESS — every architecture saturates one of the three constituents	Grounded	
Q-compute	057	Class-B exponential gap-suppression $\Gamma_{\text{fail}} \sim \Gamma_0 e^{-\xi/\xi_0}$	Grounded	
Q-compute	058	Class-C plateau $\Gamma_{\text{fail}} \rightarrow \Gamma_{\text{plateau}} > 0$ at redundancy saturation	Grounded	
Q-compute	059	Meta-architectures (EC/DD/reservoir/hybrids) = compositions that shift which constituent binds -- NO fourth class	Grounded	
Q-compute	060	M_{crit} unification: the matter-wave Q-C boundary and the qubit walls are one substrate threshold	Grounded	
QFT	013	V1 QFT-arc propagator (Kallen-Lehmann retarded form)	Grounded	
QFT	014	$N1+GR1$: V1 finite-width + 2-point function on a curved acoustic background	Grounded	
QFT	017	Free scalar Klein-Gordon ($\Box+m^2$) $\phi=0$ as the DCGT limit of V1	Grounded	
QFT	018	Non-abelian YM potential, field strength, YM equation	Grounded	
QFT	MS-II	Substrate is a $U(N)$ lattice gauge theory; escapes Nielsen-Ninomiya	Grounded	
QFT	MS-II	Spin-1/2 from relationality (tethered node carries $SU(2) \rightarrow SO(3)$ double cover)	Grounded	
QFT	MS-II	Parity violation is non-abelian: NO parity-violating abelian force	Grounded	
QFT	MS-II	Chirality γ^5 = the arrow's 'screw'	Grounded	
QFT	MS-II	$\{1,2,3\} \Leftrightarrow$ internal amplitude dimension $d=3$ (Welch-bound); no stable $SU(N \geq 4)$	Grounded	
QFT	MS-II	Three spatial dimensions because the arrow must hold committed order	Grounded	
QM-kinematics	003	Born rule $f_K = c_K ^2$ as the participation-frequency limit	Grounded	
QM-kinematics	003.5	$P_K = \sqrt{b_K} \cdot e^{i\phi_K}$ amplitude carrier, complex-valued forced	Grounded	
QM-kinematics	004.5	Sesquilinear inner-product form $\langle P, Q \rangle = \sum P_K^* Q_K$ (discrete)	Grounded	
QM-kinematics	004.6	Continuum inner-product $\int \Psi^* \Phi$ on $L^2(M, d\mu)$	Grounded	
QM-kinematics	005	Projective measurement (projector P_K , collapse, Born prob)	Grounded	
QM-kinematics	005.5	Double-slit interference $\rho = \sum P_K ^2$; which-path = V5 injection	Grounded	
QM-kinematics	006	Unitary evolution $\rightarrow$ Schrodinger $i\hbar \frac{d}{dt} \Psi = H \Psi$	Grounded	
QM-kinematics	006.5	Schrodinger equation via Stone's theorem	Grounded	
QM-kinematics	006.6	Kinetic operator $T = p^2/2m$, the factor-2	Grounded	
QM-kinematics	006.7	Schrodinger emergence in the thin-participation limit	Grounded	
QM-kinematics	007	Hilbert space = Cauchy completion of the motif algebra	Grounded	
QM-kinematics	008	Phase structure $S^1/U(1)$ on rule-types	Grounded	
QM-kinematics	008.5	Phase-independence of bandwidth = $U(1)$ gauge invariance	Grounded	
QM-kinematics	012.5	Momentum operator $p = -i\hbar \text{grad}$ = translation generator	Grounded	
QM-kinematics	012.6	Heisenberg $\Delta x \cdot \Delta p \geq \hbar/2$	Grounded	
Relativistic-QM	102	Spin-statistics $\eta = (-1)^{2s}$	Grounded	
Relativistic-QM	106 (T4)	Dirac equation $(i\gamma^\mu \partial_\mu - mc/\hbar)\Psi = 0$	Grounded	
Relativistic-QM	107	Klein-Gordon equation $(\Box + m^2 \hbar^2/c^2)\Psi = 0$	Grounded	
Relativistic-QM	108	Minimal-coupling KG + conserved 4-current	Grounded	
Relativistic-QM	109	Lorentz-representation taxonomy	Grounded	
Relativistic-QM	111	Primitive-level UV finiteness	Grounded	
Relativistic-QM	112 (T10)	Lightlike worldlines for $\sigma=0$ rule-types	Grounded	

Relativistic-QM	113	Mass structural form (Arc-M H1)	Grounded	
Relativistic-QM	114 (GRH-D1)	Existence of ≥ 1 massless Case-P gauge rule-type	Grounded	
Relativistic-QM	115	Vacuum + particle dual Fock structure ($\sigma=0$)	Grounded	
Relativistic-QM	116	MR-R / MR-P conditional massless slot	Grounded	
State-reduction	ChannelGranularity	ED's channel/pointer basis is COARSE ($\gg I_{ED}$), so bound charges are single-channel	Grounded	
State-reduction	CollapseRate	$\tau \sim \hbar/E_G$ SCALING from ED's primitives (two branch commitment-clocks $N \sim \sqrt{b}$) the single arrow cannot hold $\rightarrow$ einselection individuates)	Grounded	
State-reduction	CollapseRate	Two refinements over the rivals: a DEFINITE reduction (not Diosi's mere decoherence) + a NATIVE regulator for the point-mass divergence	Grounded	
State-reduction	CollapseRate	NOT excluded by the 2020 spontaneous-radiation bound (Lindblad jump operators = channel projectors $J=g \cdot \Pi$, FORCED from P11; committed states = exact fixed points $D[\Pi]=0$; no continuous diffusion)	Grounded	
State-reduction	OpenPiece-i	MOND enhancement of E_G (mesoscopic masses sub- a_0 in self-gravity $\rightarrow$ naive $\sim 5\text{-}50x$) is CLOSED by ED's own External Field Effect (Paper_029 horizon dipole sourced by total accel; Earth Newtonizes to $\sim 3.5e-6 \rightarrow$ ED $\sim$ Penrose-Diosi in the lab)	Grounded	
State-reduction	Positioning §3.2/§6	Locality WITHOUT superdeterminism (local commitment + A1 capacity-zero + stochastic-at-commitment; no hidden variable)	Grounded	
Substrate-evaluation	A2 (GrownNotInstalled)	A grown decoupling boundary reproduces exact channel-zero	Grounded	
Substrate-evaluation	B4/Maxwell	Integral Gauss law circulation $= 2 \cdot \pi \cdot w$ (loop-independent); the Maxwell action is LATENT in the coherence deficit ($\sin^2(d\phi/2) \sim 1/4 (grad \phi)^2$)	Grounded	
Substrate-evaluation	CleanSubstrateVector	$\gamma^5 = \text{arrow} \times \text{spontaneous spatial orientation}$; the vector clean substrate is automatically anomaly-free	Grounded	
Substrate-evaluation	GaugeStructure	Gauge group $U(N) = SU(N) \times U(1)/Z_N$ form; F^2 action; mass-gap MECHANISM [A,A] lifts flat direction iff non-abelian; single $U(1)_Y$	Grounded	
Substrate-evaluation	Gleason keystone	Channel orthogonality REDUCES to channel-distinctness given the representation; einselection primitive; Born non-contextuality; C selected	Grounded	
Substrate-evaluation	V5AttractiveSign/PhaseCoherence	V5's coupling IS the coherence content of the cross-chain superposition ($\sqrt{b_A b_B} \cos(d\phi - \text{int } A)$); MOND cross-term constructive (P12-Coh rewards alignment)	Grounded	
Substrate-evaluation	V5UnifiedBudget/QuantumDarwinism	$\rho_{loc}=1$: monogamy budget = Class-C plateau budget (one W_{max} spent two ways); per-locus live cap $N_{live} \leq W_{max}/w$	Grounded	
Black-hole	HorizonTiling	Horizon tiles ~ 1 bit/Planck cell (3 counts: holo=1, frozen 0.78, edge 0.88)	Measured	
Gravity	GR-III	Dynamical rule $b \cdot \dot{b} = D \cdot grad^2 b - \kappa \cdot \rho$: steady state $grad^2 b \sim \rho$ (0.999); $r_s \sim M$; emergent frozen $b \rightarrow 0$ horizon; area law $S \sim A$	Measured	
Gravity	GR-III	Hawking scaling $\kappa = 1/(2 r_s) \sim 1/r_h$	Measured	
Gravity	QuadraticStrain	Phase-coherence constructive sign supplied (P12-Coh rewards alignment)	Measured	
QFT	MS-II	Linking held in 3D, erasable in 4D	Measured	
QFT	MS-II	Retarded ('arrow') hopping produces net chirality signatures	Measured	
Substrate-evaluation	AreaLaw/V5UnifiedBudget	Straddling cross-chain (V5) edge count scales as AREA (r^2), reach-independent; holographic entropy = the edge count ($\sim 0.88/\text{Planck cell}$)	Measured	
Substrate-evaluation	KineticLatticeGas/CanonicalPDE	Certified continuum = kinetic lattice-gas (BALLISTIC), NOT a diffusion PDE; and does NOT generate the UDM mobility law ($\beta \sim 0$ not 2)	Measured	
Substrate-evaluation	KnotsNotCrystals/RelationalTick	No spontaneous long-range correlation-order in any sector; persistence is rhythmic (V5 binds local proper time, preserving time dilation)	Measured	
Substrate-evaluation	MassWithoutMass	Binding confines massless fronts into a sub-luminal inertial composite (mass without mass); mass(binding) $\neq$ time-dilation(commit-rate)	Measured	
Substrate-evaluation	Maxwell	The coherence-weighted continuum limit IS Coulomb, but the DETERMINATE substrate does NOT cast the smooth field (emergent shadow)	Measured	
Substrate-evaluation	PhaseCoherence/V5AttractiveSign	P12 coherence rewards alignment but stays FINITE-REACH (no crystal) 2D+3D; V5's attractive local-lock from P12's positive sign (flip destroys it)	Measured	

Tier Key				
Rank	Tier	Definition	Kind	Count
1	Standing	A settled result from another ED arc, cited here and built on — not re-derived in this folder.	Claim	7
2	Derived	Forced/proven from the 13 primitives (+ standard math) with NO paper-specific postulate — the strongest claim tier (theorems; M1).	Claim	55
3	D-via-I / Form-forced	The FORM is derived by composing inherited/standard pieces (derived-via-inherited); the form is forced, the numbers inherited.	Claim	13
4	Grounded	Structural / conditional / account-tier: form-forced GIVEN a stated input or a declared paper-specific postulate (M2/M3).	Claim	85
5	Reference	A canonical DEFINITION or catalogue (a kernel, the verdict grammar, an audit) — scaffolding, not a claim to be proven.	Input	6
6	Measured	A simulation / build-and-run / probe result on the certified substrate (or a faithful model of it).	Claim	12
7	Inherited / Consilience	Reproduces a known result/sequence framed THROUGH the substrate — supporting consilience, NOT a novel prediction.	Claim	4
8	Native / V5-conditional	Mechanism is native but rests on V5, a faithful structural addition not shown primitive-forced.	Claim	1
9	Selected/Inherited	A value taken from measurement, or a principled selection among a theorem's allowed options — not substrate-derived.	Input	41
10	Constant	A fundamental constant, by when it enters: event-scale (units, present at the first events) / derived-combination / inherited-late.	Input	10
11	Primitive	One of the 13 postulated axioms (P01-P13) — the corpus's single 'conditional on'; postulated, not derived, not claimed minimal.	Input	13
12	Postulated	Rests on a named paper-specific postulate (a declared commitment BEYOND the 13 primitives).	Input	24
13	Asserted	An A->position / A->assertion: stated in the paper's own audit as a position, NOT proven (the weakest claim tier).	Claim	16
14	Prediction	A falsifiable, experiment-facing bet — carries a kill-condition (see the Falsifier / Kill-Cond column).	Claim	23
15	Identification	A structural mapping onto an existing EXTERNAL framework (e.g. Penrose CCC, Jacobson's equation of state).	Claim	3
16	Wall	A proven or honestly-checked LIMIT — the order or result ED does NOT reach (a load-bearing 'no').	Claim	4
17	Synthesis	An arc consolidation / overview / organizing lens; composes standing results, adds no new derivation.	Meta	30
18	Superseded	Retracted / withdrawn / superseded by a later corpus result (kept for provenance).	Meta	3
19	Open	A declared OPEN / unclosed item — a named gap or frontier.	Meta	51